
Sunghwan Ko

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RESEARCH INTERESTS

- Coxeter polytopes and reflection groups
- Convex real projective structures and their deformation spaces
- Hilbert geometry and higher Teichmüller theory
- Discrete subgroups of Lie groups

EDUCATION

Seoul National University

Ph.D. in Mathematics

Advisor: Gye-Seon Lee

Mar 2021-Feb 2027

(expected)

Sogang University

B.Sc. in Mathematics

Mar 2015-Feb 2021

RESEARCH PAPERS

PREPRINT

- Finiteness of integral representations on 2-perfect truncation polytopes. [arXiv:2604.22243](https://arxiv.org/abs/2604.22243), 2026

WORKS IN PROGRESS

- Infinitesimal deformations of reflection groups and their orientation-preserving subgroups. With Eun-Gyu Jang
- Integral representations on compact Coxeter polygons. With Seong Youn Kim

TALKS

Workshop for Young Mathematicians in Korea

Finiteness of integral representations on 2-perfect truncation polytopes · 19 Aug 2026

3rd KIAS Graduate Student Meeting in Pure Mathematics, KIAS

Infinitesimal deformations of representations and group cohomology · 7 May 2026

The 21st East Asian Conference on Geometric Topology, Ewha Womans University

Finiteness of integral Vinberg representations for Coxeter truncation polytopes · 4 Feb 2026

Dynamical Group Theory VII, Seoul National University (Lightning talk)

Finiteness of integral Vinberg representations for truncation polytopes · 26 Nov 2025

Exchange Seminar, Seoul National University

Finiteness of integral representations arising from Coxeter truncation polytopes · 13 Nov 2025

Séminaire Groupes et Actions, Laboratoire de Mathématiques d'Orsay

Finiteness of integral representations coming from truncation polytopes · 12 May 2025

Exchange Seminar, Seoul National University

Finiteness of integral representations of Coxeter 4-gons · 26 Sep 2023

SELECTED CONFERENCES AND WORKSHOPS

- Workshop for Young Mathematicians in Korea, KAIST, 19-21 Aug 2026
- 3rd KIAS Graduate Student Meeting in Pure Mathematics, KIAS, 7 May 2026
- Homogeneous Dynamics and Anosov Representations, Simons Laufer Mathematical Sciences Institute, virtual, 20-24 Apr 2026
- Teichmüller Theory and Related Topics, Fillmore Grand Binhu, Hefei, 4-10 Apr 2026
- The 21st East Asian Conference on Geometric Topology, Ewha Womans University, 2-5 Feb 2026

SELECTED CONFERENCES AND WORKSHOPS - CONTINUED

- Introductory Workshop: Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups, Simons Laufer Mathematical Sciences Institute, 26-30 Jan 2026
- Pathways Workshop: Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups, Simons Laufer Mathematical Sciences Institute, 21-23 Jan 2026
- Dynamical Group Theory VII, Seoul National University, 26-28 Nov 2025
- One-Day Festival of Group Actions, Geometry and Dynamics, KIAS, Seoul, 6 Aug 2025
- KAIST-KIAS Geometry and Topology Exploration Workshop I, KIAS, 10-13 Jun 2025
- Higher Rank Geometric Structures, Higgs Bundles and Physics, IHP, Paris, 19-27 May 2025
- Geometric Structures and Discrete Group Actions, Higher Rank Geometric Structures program, 14 Apr-11 Jul 2025
- Teichmüller Theory and Beyond, Seoul National University, 17-21 Feb 2025
- International Workshop on Higher Teichmüller Theory, Chern Institute of Mathematics, Tianjin, 15-24 May 2024

TEACHING EXPERIENCE

Seoul National University	
Topics in Topology	Fall 2026
Introduction to Differential Geometry	Spring 2026
International Calculus Practice II	Fall 2025
International Introduction to Linear Algebra	Spring 2025
International Introduction to Linear Algebra	Fall 2024
International Calculus Practice I	Spring 2024
International Calculus Practice II	Fall 2023
International Calculus Practice I	Spring 2023
International Introduction to Differential Geometry II	Fall 2022
International Calculus Practice I	Spring 2022
International Calculus Practice II	Fall 2021
International Calculus Practice I	Spring 2021

AWARDS AND HONORS

Outstanding TA (Calculus I), Seoul National University	Spring 2021
Dean's List, Sogang University	Feb 2021
Bronze Award, Korean Contest of Mathematics for Undergraduate Students, Korean Mathematical Society	Dec 2018

MILITARY SERVICE

Completed mandatory military service, Republic of Korea Army · Apr 2017-Dec 2018